Pressure in a Moving Fluid

In a moving fluid, the hydrostatic pressure formula is not accurate in finding the actual pressure of a moving fluid at a certain point.

Stagnation Pressure

The pressure of a moving fluid can be thought of as two components, the static and dynamic pressure.

Static Pressure

Static pressure is the pressure the liquid exerts only due to the liquid and not its motion.

In some cases, static pressure is equal to hydrostatic pressure, for example when the fluid is travelling horizontally, however generally static pressure is a measured quantity.

Static pressure can be thought of as the “potential energy” of a fluid.

Dynamic Pressure

Dynamic pressure is not an actual kind of pressure, but more of a representation of the kinetic energy of a fluid.

Dynamic pressure is given as:

$$P = \frac{1}{2}\rho U^2$$

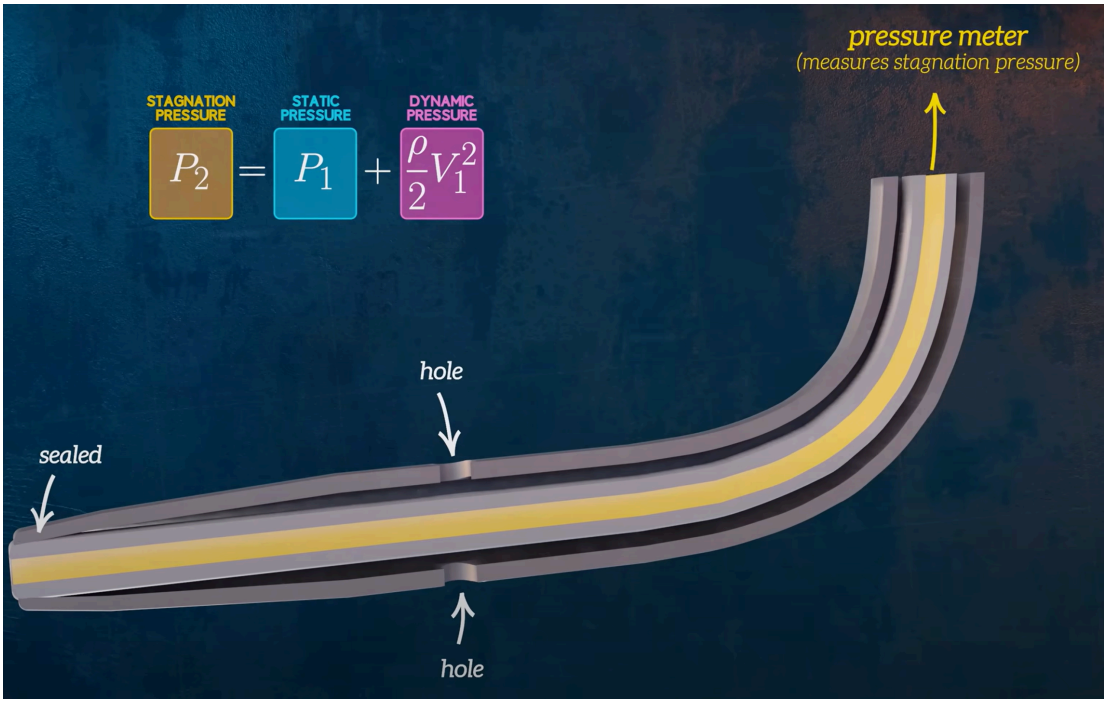
where U is the fluid velocity.

Stagnation Pressure is essentially the total pressure of a moving fluid, which is the sum of the two pressure components:

$$P_{\text{stagnation}} = P_{\text{static}} + P_{\text{dynamic}}$$

▼ Pitot Tube Example

A pitot tube is used to measure the fluid velocity of a fluid by immersing it into a moving fluid, pointed in the direction of the flow velocity.

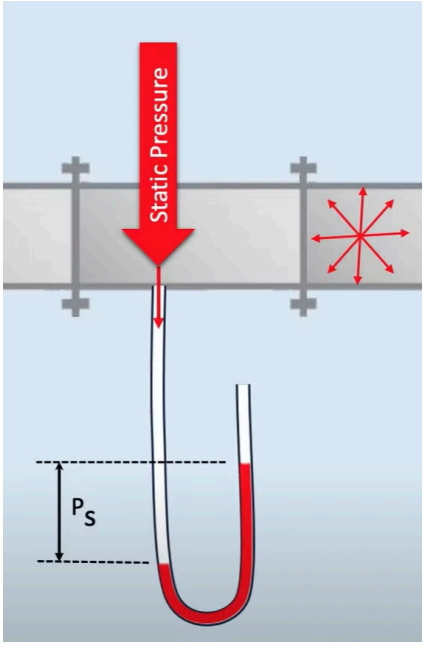


The pitot tube consists of two kinds of holes, one in the same direction as the flow direction, and another perpendicular to the flow direction. Both holes aim the fluid towards two separate fluid meters.

Consider the tubing in the same direction as the flow direction. As the hole is in the same direction, fluid flows in due to its static pressure pushing it towards the hole and the dynamic pressure due to the velocity towards it.

When fluid reaches and flows into the tube, it is stopped by the bend and thus all of its dynamic pressure is converted to static pressure. Thus, the pressure meter will measure the original static pressure AND the static pressure from the converted dynamic pressure, aka stagnation pressure.

For the holes aimed perpendicularly, the fluid flowing through will not have any dynamic pressure, as it does not flow in with the fluid velocity. Instead, fluid flows into these holes due to the fluid pushing in all directions, which happens to flow into these holes.



As such, the pressure captured by the pressure meter is only due to its original static pressure.

One thing to note is that since the fluid is flowing horizontally, there is no fluid flowing vertically, meaning that static pressure downwards (which is the same in all directions, however if we consider downwards) is also the pressure due to gravity, aka hydrostatic pressure.

Thus, with static and stagnation pressure, dynamic pressure and thus the fluid velocity can be calculated:

$$P_{\text{dynamic}} = P_{\text{stagnation}} - P_{\text{static}}$$
$$\frac{1}{2}\rho U^2 = P_{\text{stagnation}} - P_{\text{static}}$$
$$U = \sqrt{\frac{2\left(P_{\text{stagnation}} - P_{\text{static}}\right)}{\rho}}$$

Head

When static, dynamic or stagnation pressure is expressed in liquid column units, we call it static head, dynamic head or stagnation head respectively.

Vapour Pressure

Vapour Pressure is a threshold of the liquid such that when it drops BELOW the vapour pressure, the liquid boils.

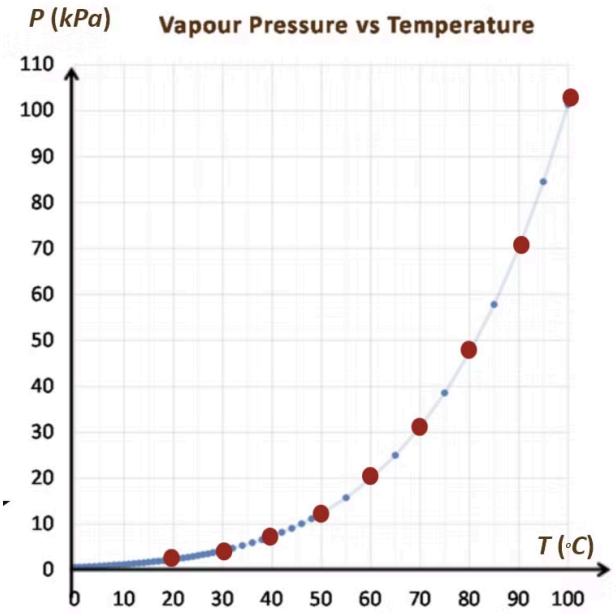
If we consider a liquid column exposed to some air, we know that its hydrostatic pressure is $P_0 + \rho g h$. When atmospheric pressure is low enough, the intermolecular forces of the liquid are strong enough to overcome the less strong atmospheric pressure (air presses less on liquid), causing it to boil.

Thus, when a liquid has a lower pressure, it is more likely to boil (once it drops below vapour pressure)

Vapour pressure is generally proportional to temperature (higher temperature, higher vapour pressure), generally with a lookup table.

Notice from the graph below, that it is harder to boil a fluid at low temperatures than at higher temperatures due to very low pressures needed.

e.g. boiling at 50C requires less than 10 kPa, which is harder to reach than 110 kPa when boiling at 100C. (more range from 0 to vapour pressure)



Vapour Lock

Vapour Pressure is very important as a fluid may boil at certain locations (called local boiling) in liquid flow systems (e.g. piping) when pressure drops below the vapour pressure, or when temperature is too high,

causing a higher vapour pressure, moving the threshold too high for the liquid's pressure.

This causes bubbles of vapour to form, which could block the flow in these systems. This condition of vapour bubbles forming and blocking flow is called vapour lock.

To prevent this, pressure of the fluid must be maintained above the vapour pressure at all points along the flow system, or the temperature must be maintained low enough to keep the vapour pressure low.

Cavitation

The formation of vapour bubbles due to local boiling may cause cavitation.

Cavitation occurs when low pressure vapour bubbles (as they were formed in low pressure) are swept to other parts of the flow system with high pressure. The extreme pressure difference causes the bubbles to collapse, generating destructive pressure waves.

Cavitation reduces performance, generated unwanted vibrations and noise, and causes damage to equipment. Over a long period of time, multiple cavitations can cause erosion, surface pitting and fatigue damage